

$$\begin{aligned}
& C_{\phi} \frac{(21)}{\partial^2} \rightarrow \\
& \frac{1}{16\pi^2} \left(\frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 \text{LF}_{3,2,0}[m_1, \tilde{\mu}] - \frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 \text{LF}_{3,3,-1}[m_1, \tilde{\mu}] - \frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 \text{LF}_{4,2,-1}[m_1, \tilde{\mu}] + \right. \\
& \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 \text{LF}_{3,2,0}[m_2, \tilde{\mu}] - \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 \text{LF}_{3,3,-1}[m_2, \tilde{\mu}] - \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 \text{LF}_{4,2,-1}[m_2, \tilde{\mu}] - \\
& \frac{1}{8} g_1^4 m_1^2 \tilde{\mu}^2 \text{LF}_{4,2,-1}[\tilde{\mu}, m_1] + \frac{3}{8} g_2^4 m_2^2 \tilde{\mu}^2 \text{LF}_{4,2,-1}[\tilde{\mu}, m_2] - \\
& 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 \text{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_1] + 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 \text{LF}_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& \frac{1}{2} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 \text{LF}_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] + 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 \text{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{p}}, m_q^{\text{s}}] + \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{p}}, m_q^{\text{s}}] + \\
& \frac{3}{4} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_d^{\text{r}}, m_q^{\text{p}}, m_d^{\text{t}}, m_q^{\text{s}}] + \frac{3}{4} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \\
& \text{LF}_{2,2,1,1,-1}[m_d^{\text{r}}, m_q^{\text{s}}, m_d^{\text{t}}, m_q^{\text{p}}] - \frac{1}{2} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{p}}, m_l^{\text{s}}] + \\
& \frac{1}{2} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{p}}, m_l^{\text{s}}] + \frac{1}{4} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \\
& \text{LF}_{2,2,1,1,-1}[m_e^{\text{r}}, m_l^{\text{p}}, m_e^{\text{t}}, m_l^{\text{s}}] + \frac{1}{4} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_e^{\text{r}}, m_l^{\text{s}}, m_e^{\text{t}}, m_l^{\text{p}}] + \\
& \frac{1}{8} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_l^{\text{p}}, m_l^{\text{s}}, m_e^{\text{r}}, m_e^{\text{t}}] + \frac{3}{8} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \\
& \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}, m_d^{\text{t}}] - \frac{9}{4} \tilde{\mu}^2 y_d^{\text{sr}} y_u^{\text{pt}} \overline{a_d^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{4} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] - \frac{3}{4} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \\
& \text{LF}_{3,1,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] - \frac{3}{8} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_u^{\text{r}}, m_q^{\text{s}}, m_u^{\text{t}}] - \\
& \frac{3}{8} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_u^{\text{t}}, m_q^{\text{s}}, m_u^{\text{r}}] - \\
& \left. \frac{3}{4} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_u^{\text{r}}, m_u^{\text{t}}, m_q^{\text{p}}, m_q^{\text{s}}] \right)
\end{aligned}$$